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Advances in Space Research 74 (2024) 1576–1585

ADVANCES IN SPACE RESEARCH
(a COSPAR publication)

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Spatial-temporal variation of water vapor scale height and its impact factors in different climate zones of China

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Received 3 February 2024; received in revised form 21 April 2024; accepted 8 May 2024
Available online 13 May 2024

Abstract

Water vapor scale height reflects the vertical distribution of water vapor in the atmosphere, which is a vital parameter in tropospheric zenith wet delay (ZWD) modeling and GNSS meteorology. This study accurately calculated the long-term water vapor scale height using 89 radiosonde stations in China, analyzed the characteristics in temporal and spatial variations and explored the impact factors of the water vapor scale height from the perspective of the climate zones. The numerical results indicate that the water vapor scale height in China exhibits annual and semiannual variations with a rising-then-decreasing trend across the four seasons. A relatively larger value for the water vapor scale height were observed in the subtropical monsoon climate and the temperate continental climate zone, in which the values are greater than 2.35 km, while the values in temperate monsoon climate and the alpine plateau climate zone are relatively small. In addition, the water vapor scale height in different climate zones shows different trends, with a downward trend appeared in the subtropical monsoon climate and the alpine plateau climate zone, while an upward trend was observed in the temperate monsoon climate and the temperate continental climate zone. Furthermore, a positive correlation between water vapor scale height and temperature, pressure, and atmospheric precipitable water was found in China, with a varied degree of correlation in different climate zones.

Keywords: GNSS Meteorology; Water vapor scale height; Radiosonde; Climate zones

1. Introduction

As the most prevalent greenhouse gas in Earth's atmosphere, water vapor plays a pivotal role in shaping our planet's climate and maintaining a temperature range conducive to life (Jacob, 2001; Li and Long, 2020). It exerts a profound influence on climate change and has multifaceted impacts on our environment (Dessler et al., 2008). One of its key roles lies in the latent heat it releases as it circulates through the atmosphere and flows horizontally,

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https://doi.org/10.1016/j.asr.2024.05.019

which is a critical component of the energy balance in both the northern and southern hemispheres (Bevis et al., 1992). Additionally, the phase change of water vapor significantly influences the stability of the atmospheric structure, directly contributing to the formation of severe convective weather and the occurrence of destructive weather events (Yang et al., 2022a). The water vapor also serves as a medium for signal propagation, and it has been shown to directly impact the propagation of electromagnetic wave signals in Earth's satellite observation systems (Chen, 1998). This influence extends to various services such as radio communication, navigation, and positioning, underscoring its relevance in our daily lives (Yao and Zhao, 2022).

The troposphere contains an astounding 99 % of the Earth's water vapor, with a significant concentration found at heights below 2 km, constituting approximately half of the total water vapor content (Li et al., 2008). Despite its relatively low abundance in the atmosphere, water vapor is one of the most dynamic and variable components, exhibiting fluctuations both in time and space (Rocken et al., 1997; Yang et al., 2022b). The water vapor scale height represents the height increment for the water vapor density to decrease to 36.7 % ($1/e$) of its value at a certain height (Tomasi, 1981; Reitan, 1963). Meteorological sounding, a widely employed technique for studying the vertical distribution of water vapor, involves the deployment of water vapor detection instruments on sounding balloons or manned aircraft at different altitudes (Sun et al., 2023; Miloshevich et al., 2006; Ferrare et al., 1995). An in-depth exploration of water vapor scale height enhances our comprehension of atmospheric water vapor distribution and variability. At the same time, the vertical distribution of water vapor is related to the atmospheric turbulent structure, and by observing the vertical distribution of water vapor, the interaction between water vapor and turbulent structure in the atmospheric boundary layer can be better investigated, so as to gain a deeper understanding of the mechanism of water vapor transport and change in the atmosphere (Ruf and Beus, 1997). The water vapor scale height can also provide valuable constraint information for global climate models, helping them to more accurately predict the behavior of atmospheric boundary layer water vapor (Schüler, 2014). Additionally, it plays a pivotal role in GNSS Meteorology. In the precise correction of zenith wet delay (ZWD) for signals from global navigation satellite systems traversing the troposphere, it is assumed that ZWD follows an exponential change vertically and the water vapor scale height was served as the elevation correction coefficient (Schüler, 2014; Huang et al., 2023a, 2023b). The accuracy of the water vapor scale height is not only helpful for understanding the vertical variation of ZWD, but also contributes to refine the accuracy of the ZWD models (Yao et al., 2015; Yang et al., 2021; Zhao et al., 2023a, 2023b; Li et al., 2024). Furthermore, water vapor scale height is also needed to characterize and constrain the vertical distribution of the wet refractive index (water vapor density) in the construction of vertical constraints in GNSS tomography (Flores et al., 2001; Yang et al., 2023a, 2023b).

Based on the clear-sky sounding data in the Po Valley zone of Italy, Tomasi (1977) calculated the water vapor scale height in the zone and found that it was small in summer and large in winter, with an overall change of 1.68–1.92 km. Using the meteorological data of 12 sounding stations in Xinjiang for 10 years, Zhang (2002) obtained that the average water vapor scale height in this zone was 2.17 km. Based on the clear-sky observation data of atmospheric water vapor content and surface water vapor density from 2002 to 2006, Li et al. (2008) retrieved the water vapor scale height in Hefei and summarized its seasonal and diurnal variation characteristics. While these studies provided valuable insights into specific urban or zonal areas, it is essential to acknowledge that water vapor distribution is subject to rapid changes. Thus, research focusing solely on localized areas may not capture the broader variations in water vapor scale height. Zhang et al. (2015) attempted to used ECMWF reanalysis data to construct an empirical global model for the water vapor scale height, in which only the annual and semiannual amplitude were estimated. The spatial characteristics of water vapor scale height and their temporal analysis were still not sufficient.

In this paper, the radiosonde data which truly measured the atmospheric state of vertical layers, is utilized to calculate the water vapor scale height in China, a comprehensive analysis of the temporal and spatial variation characteristics and patterns was conducted for the water vapor scale height from a new perspective of climate zones, and the its impact factors were also explored.

2. Data and methods

2.1. Data

In this study, the daily data of 89 radiosonde stations in China are collected from the University of Wyoming Weather Data website. These data include meteorological parameters such as temperature, pressure, and relative humidity, covering the range from the Earth's surface to an altitude of approximately 30 km. Note that the radiosonde data are collected at two different times, i.e., UTC 12:00 and UTC 0:00. The different climate types in China were considered to conducte the spatiotemperal and impact analysis, which is shown in Fig. 1 including the temperate continental climate zone, the alpine plateau climate zone, the temperate monsoon climate zone, and the subtropical monsoon climate zone.

2.2. Calculation of water vapor scale height

The distribution of water vapor density in vertical can be expressed as a negative exponential function as follows:

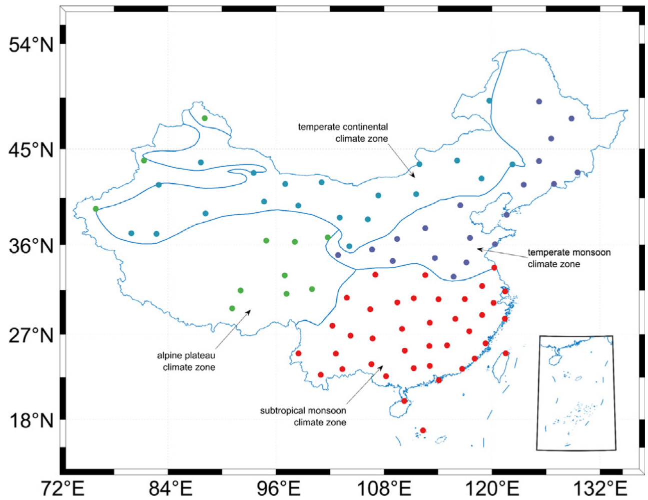

Fig. 1. The climate zones in China and the geographical distribution of radiosonde stations.

$$\rho_z = \rho_0 * e^{-z/H} \tag{1}$$

where ρ_0 is the surface water vapor density, ρ_z is the density of water vapor at a height of z, and H is the water vapor scale height, respectively. The Integrated water vapor (IWV) is the integral of the water vapor density ρ along the whole atmospheric column. Thus, according to formula (1) can conclude that:

$$IWV = \int_0^z \rho_z dz = \int_0^z \rho_0 * e^{-z/H} dz = (1 - e^{-z/H}) * \rho_0 H \tag{2}$$

where z is the height at which the water vapor density approaches 0 zero, thus $e^{-z/H} \to 0$. According to formula (2), the water vapor scale height H can be calculated as follows (Li et al., 2008):

$$H = \frac{IWV}{\rho_0} \tag{3}$$

The surface vapor concentration can be calculated by the following formula:

$$\rho_0 = \frac{es}{R_v * T} \tag{4}$$

where es represents water vapor pressure, T is temperature and $R_v = 461.5 J/(kg * K)$ denotes a gas constant. In addition, IWV can be converted by atmospheric precipitable water, as follows:

$$IWV = \rho_w * PWV \tag{5}$$

where $\rho_w \approx 999 kg/m^3$ means the liquid water density and the PWV (Precipitable Water Vapor) can be calculated accurately by the following formula based on the stratified meteorological data of radiosonde stations:

$$PWV = \frac{1}{\rho_w * g} * \sum_{i=1}^{n} (sh(i) * pre(i)) \tag{6}$$

where $g \approx 9.80665 m/s^2$ is the gravitational acceleration, i denotes the i th pressure level and n is the total number of layers. $sh(i)$ and $pre(i)$ represent the specific humidity and pressure of the i th pressure level, respectively. The specific humidity can be calculated by the following formula:

$$sh = \frac{0.622 * es}{(pre - 0.378 * es)} \tag{7}$$

where the water vapor pressure can be calculated by the modified Magnus formula:

$$\begin{cases} es = 6.1078 * 10^{\left(7.5 * \frac{T}{(237.3+T)}\right)} * \frac{rh}{100}, T > 0 \\ es = 6.1078 * 10^{\left(9.5 * \frac{T}{(265.5+T)}\right)} * \frac{rh}{100}, T \leqslant 0 \end{cases} \tag{8}$$

where rh represents relative humidity.

2.3. Analysis method

In order to obtain the change trend of water vapor scale height in different stations and zones, the least square linear fitting method is used as follows:

$$H = p_1 * doy + p_2 \tag{9}$$

where doy is the day of the year, p_1 is the change rate of water vapor scale height and p_2 is the fitting coefficient.

To detect the periodicity of water vapor scale height, the power spectral density (PSD) of water vapor scale height time series is obtained by periodogram method as follows (Vlachos et al., 2005):

$$P\left(f_{k/N}\right) = \left\| X\left(f_{\frac{k}{N}}\right) \right\|^2 \qquad k = 0, 1 \ldots \left[\frac{N-1}{2}\right] \tag{10}$$

where X is the discrete Fourier transform of the time series and the subscript k/N represents the frequency captured by each coefficient. The periodogram is obtained by the square of the length of each Fourier series.

To further determine the periodicity of water vapor scale height, the annual and semi-annual cycles of water vapor scale height are modeled in the form of trigonometric functions. The form is as follows (Zhang et al., 2015):

$$\begin{aligned} H = a_0 + a_1 * \cos\left(2\pi * \frac{doy}{365.25}\right) + a_2 \\ * \sin\left(2\pi * \frac{doy}{365.25}\right) + a_3 * \cos\left(4\pi * \frac{doy}{365.25}\right) \\ + a_4 * \sin\left(4\pi * \frac{doy}{365.25}\right) \end{aligned} \tag{11}$$

where a_0, a_1, a_2, a_3, a_4 is the model coefficient.

3. Analysis and discussion

The annual average of the water vapor scale height at 89 sounding stations in China from 2016 to 2019 are counted and shown in Fig. 2. It can be seen that the distribution of the water vapor scale height in China varied with a range from 1.79 km to 3.02 km. The largest annual average value

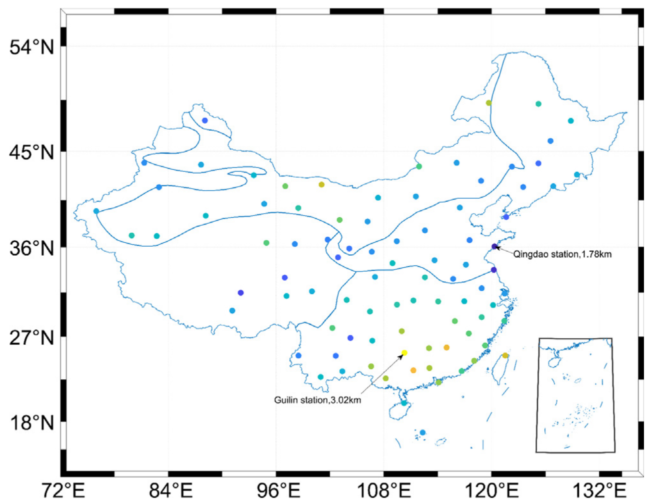

Fig. 2. Distribution of the annual average of water vapor scale height in different climate zones.

is appeared in the subtropical monsoon climate zone with a value more than 2.45 km. Note that there are also some stations with large annual average value in the temperate continental climate zone, for example, the annual average value reached to 2.71 km at Ejin Banner station. Across the whole country, the largest and smallest annual average values for the water vapor scale height are the Guilin station and Qingdao station with a value of 3.02 km and 1.78 km, respectively. The mean values are 2.06 km, 1.98 km, 2.15 km, and 1.84 km for the temperate continental, the alpine plateau, the temperate monsoon, and the subtropical monsoon climate zone, respectivly.

To further explore the differences of water vapor scale height in each climate zone, the statistical values including median, mean, maximum and minimum values were shown in Table 1. The largest median and mean values appeared in the subtropical monsoon climate zone, followed by the temperate continental climate zone, and the difference of these two values is very small for the other two climate zones. Note that the maximum and minimum values are 2.71 and 1.78 km, which are appeared in the temperate continental and the temperate monsoon climate zone, respectively. The difference between the maximum and minimum values of the subtropical monsoon climate zone is the largest reaching 0.78 km. When using 2.34 km as a reference which is the average value of the four climate zones, the percentage of stations with water vapor scale height greater than the reference value are 76.3 %, 15.8 %, 50.0 % and 18.2 % for the four climate zones, respectively.

The monthly water vapor scale height was computed in each climate zone as well as in the whole country, which were shown in Fig. 3. It can be seen that the water vapor scale height of the national zone has a trend of first rising and then falling, and the largest value is 2.42 km and appeared in July. In this month, the specific values were 2.49, 2.29, 2.40 and 2.39 km for the four climate zones, respectively. On the other hand, the water vapor scale height of the national zones in December has the lowest value of 2.04 km, and the specific values for the four different climate zones are 2.06, 1.97, 2.15 and 1.84 km, respectively. Note that the values for all months are greater than 2 km in the subtropical monsoon and the temperate continental climate zone, and a value greater than 2.5 km is observed in June of the subtropical monsoon climate zone. From October to February, these values are all at a low range in each climate zone, especially in the temperate monsoon and the alpine plateau climate zone which has a value smaller than 2.0 km. It is observed that the lowest value is 1.83 km that occurred in December of the alpine plateau climate zone. Overall, the water vapor scale height of the temperate continental and the subtropical monsoon climate zone in each month are higher than those in the alpine plateau and the temperate monsoon climate zone. It can also be observed that the water vapor scale height of all zones exhibits seasonal fluctuations.

To illustrate the differences of the water vapor scale height in four seasons, namely, spring (March to May), summer (June to August), autumn (September to November), and winter (December to February), these values in each climate zone were counted, as shown in Fig. 4 the form of boxplot. The seasonal differences in water vapor scale height were easily observed with the median and average values of 2.32, 2.50, 2.22 and 2.27 km and 2.34, 2.47, 2.27 and 2.28 km for spring, summer, autumn and winter, respectively. The average and median values in summer are higher than those in other seasons, since the strong heating of the ground by summer sunlight causes extensive convective motion in the lower atmosphere. Furthermore, the seasonal differences in water vapor scale height varies among different climate zones, with the most significant being in the temperate monsoon climate zone and the least significant being in the subtropical monsoon climate zone. The median values of these two climate zones are 2.48, 2.49, 2.46 and 2.45 km for the four seasons, respectively.

Table 1
Statistics of water vapor scale height in different climate zones.

zone	median (km)	mean (km)	maximum (km)	minimum (km)	>2.34 km (%)
subtropical monsoon climate zone	2.45	2.45	2.62	1.84	76.3
temperate monsoon climate zone	2.18	2.19	2.44	1.78	15.8
temperate continental climate zone	2.33	2.36	2.71	2.09	50.0
alpine plateau climate zone	2.17	2.20	2.54	1.92	18.2

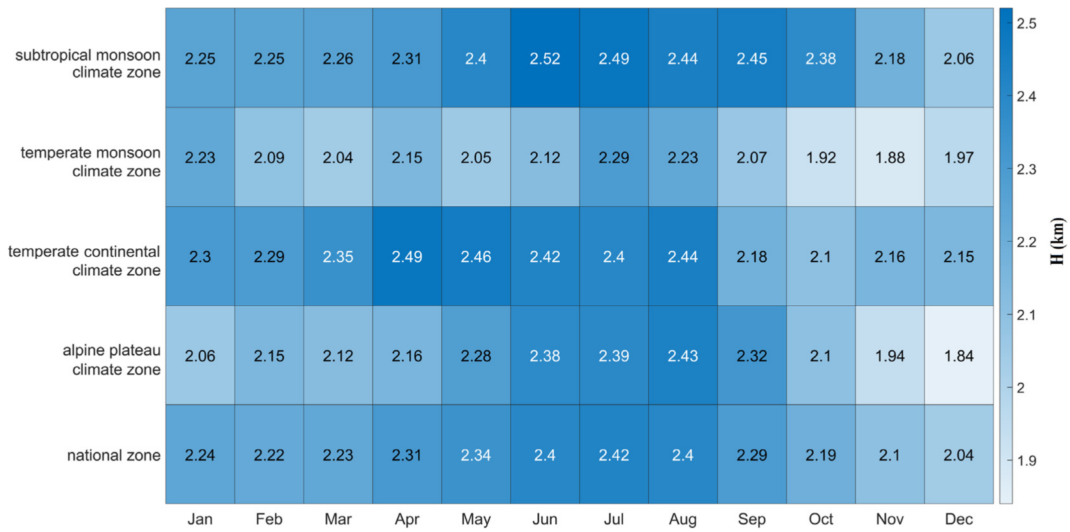

Fig. 3. Comparison of monthly water vapor scale height data in different climate zones.

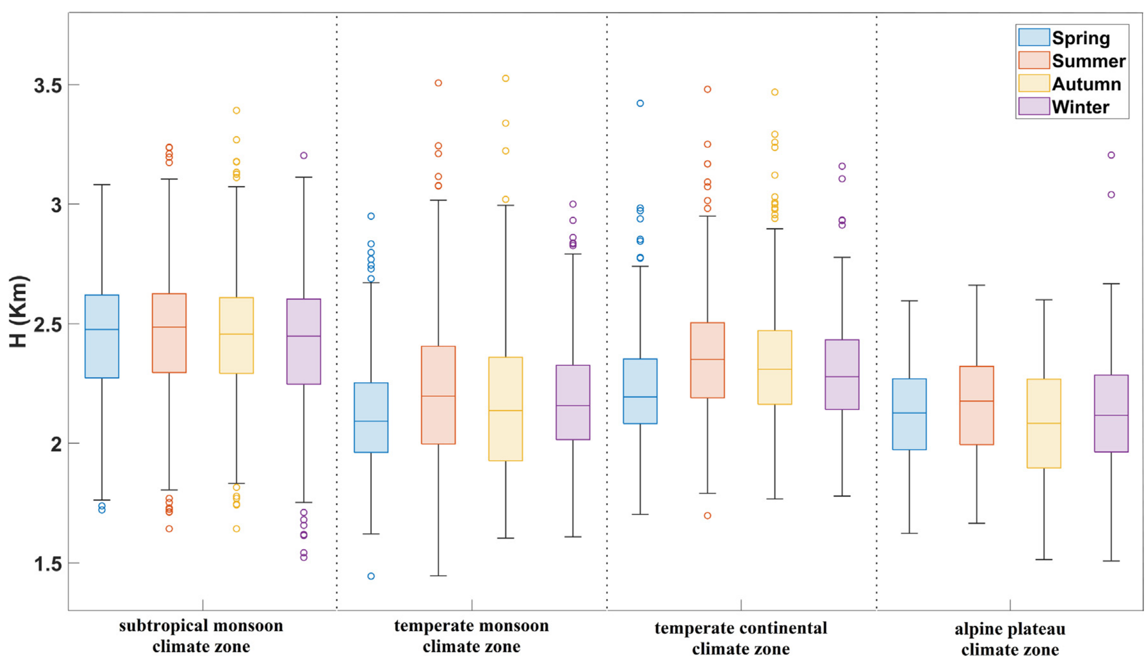

Fig. 4. Boxplot of water vapor scale height in different seasons.

To show the water vapor scale height at different moments, these values at UTC 12:00 and 0:00 were counted in each climate zone, and their histograms were shown in Fig. 5. The water vapor scale height in each histogram well fitted the normal distribution, namely, with most of values concentrated on the mean and gradually decrease to sides. It can be seen that the water vapor scale height at UCT 12:00 is greater than that at UTC 0:00 in each climate zone, especially for the temperate continental and the alpine plateau climate zone with the mean values are 2.37/2.24 km and 2.30/2.08 km for UTC 12:00 and 0:00, respectively. Note that the water vapor scale height is more stable at UTC 12:00 than that at UTC 0:00 with a smaller σ at this epoch in each climate zone.

The water vapor scale height of all stations from 2016 to 2019 were utilized to conduct the least square fitting, and the trend of each station were plotted in Fig. 6, in which the reds and greens represent the increase and decrease trend, respectively. Note that there is a strong correlation between the trend and climate zone: in the subtropical monsoon and alpine plateau climate zone, the water vapor scale height exhibits a trend of decline, whereas in the temperate monsoon and temperate continental climate zone, it shows an upward trend. Specifically, 81.82 % and 71.05 % of stations in the alpine plateau and the subtropical monsoon climate zone show the decline trend respectively, and 57.89 % of stations in the temperate monsoon climate zone exhibit the trend of increase. Among all stations, the

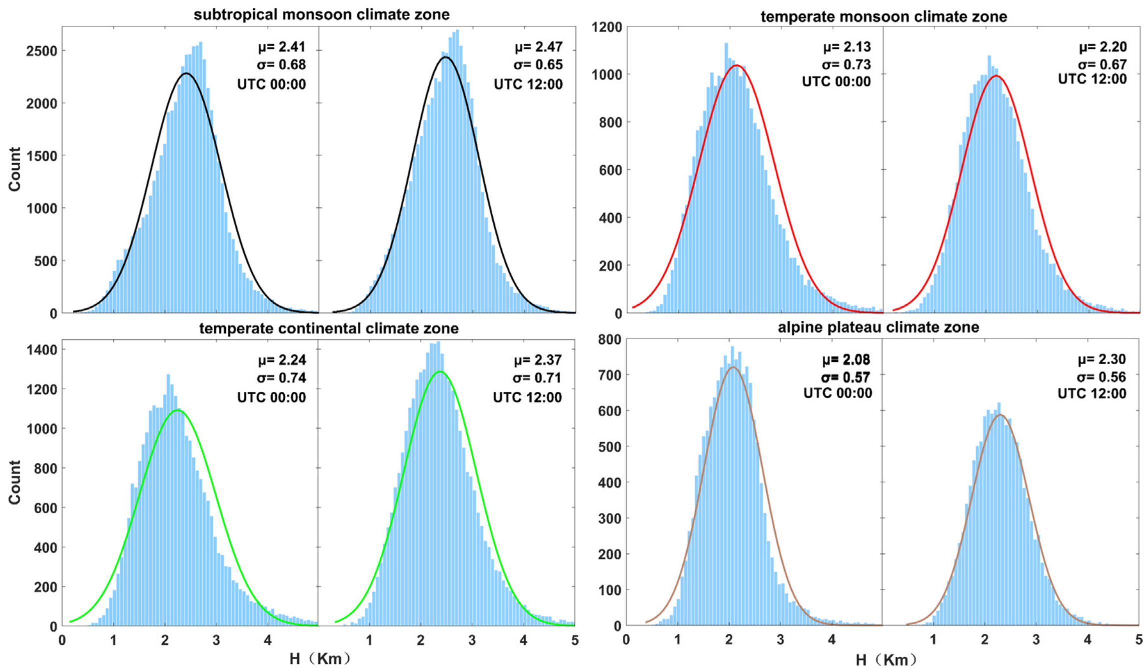

Fig. 5. Histogram of water vapor scale height distribution at UTC 12:00 and 0:00 in different climate zones.

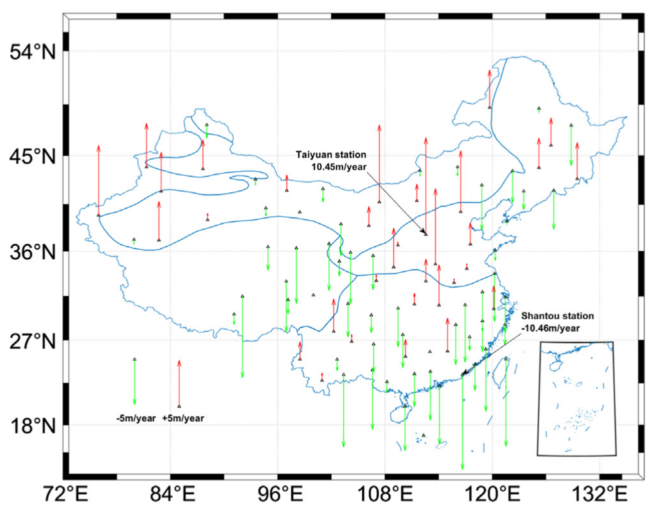

Fig. 6. Variation trend of water vapor scale height in different climate zones.

Shantou station represents the largest decline of −10.46 m per year, and the Taiyuan station has an annual increase of 10.45 m as the with the largest rise, conversely. Nationally, the mean change is 3.38 m per year, but regardless of whether it is increasing or declining, the overall magnitude of change remains relatively small, suggesting that the water vapor scale height in China exhibits relatively stable conditions.

To determine the periodicity of the water vapor scale height, the spectral analysis is conducted in each station and four sites were selected as representatives from the different climate zones to show their power spectral density-period diagram in Fig. 7. It exhibits an obvious surge and peak at a period of approximately one year in each site, which indicates a significant annual periodicity in water vapor scale height. Furthermore, the sharp increase also appears at a period of nearly 0.5 years at some stations, indicating a semi-annual periodicity. Another thing to note is that the semi-annual cycle is even more obvious than the annual cycle in some sites like Beijing.

Based on the periodic characteristics of water vapor scale height determined through spectral analysis, Eq. (9) was applied to fit the annual and semiannual cycle in different climate zones, and the time series and their corresponding fitting curves were plotted in Fig. 8 for the four climate zones. The annual and semiannual variation in different climate zones were clearly showed by the red curves. The maximum amplitude of annual cycle is 0.18 km in the alpine plateau climate zone, while the minimum value is 0.01 km in the temperate monsoon climate zone. In addition, obvious semi-annual cycle of the water vapor scale height was observed in the temperate monsoon and the alpine plateau climate zone with values of 0.15 km and 0.12 km, respectively.

Furthermore, the annual and semiannual cycle was fitted in each station and their distributions of the amplitude were shown in Fig. 9. It is clearly that the annual amplitude is greater than the semiannual amplitude in each site, and their average values are 0.20 km and 0.09 km, respectively. The largest and smallest annual amplitude of the water vapor scale height appeared at Kunming and Nanchang station with a value of 0.51 km and 0.03 km, respectively. For the semi-annual amplitude, Hailar and Weining station exhibit maximum and minimum values with 0.43 km and 0.01 km, respectively. The differences in the amplitudes of the annual and semiannual cycle can also be observed

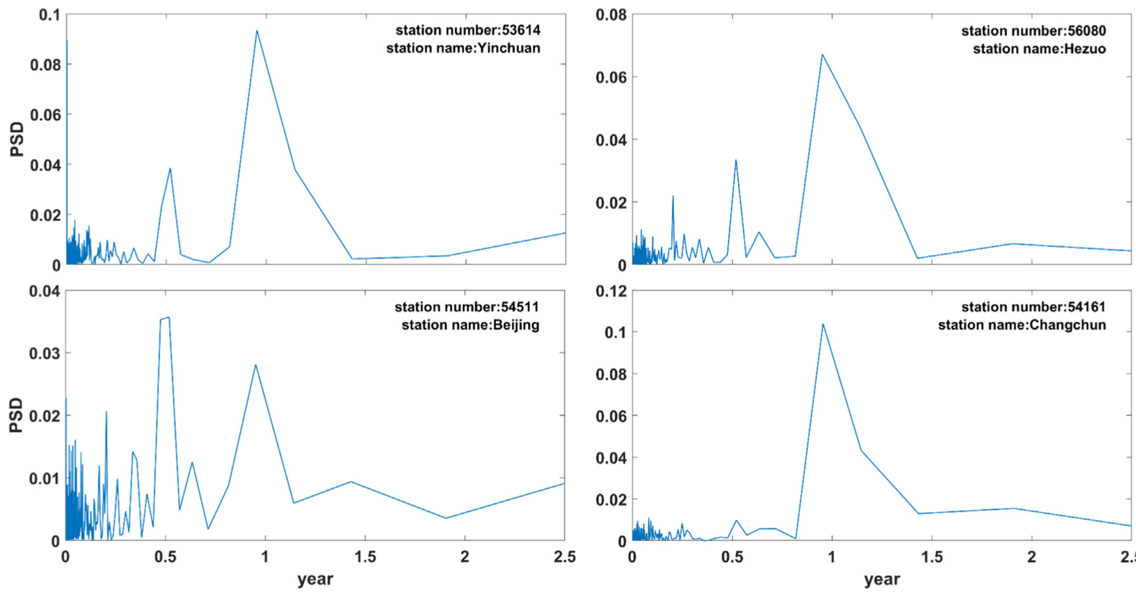

Fig. 7. Power Spectral Density – period diagram of the four selected stations.

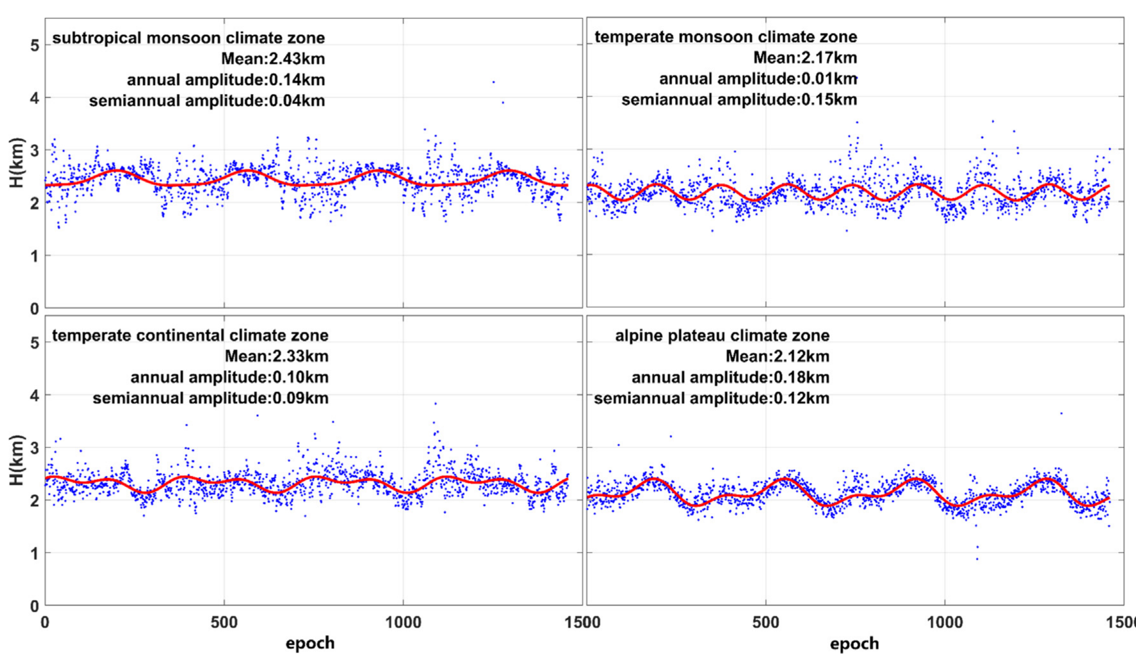

Fig. 8. Fitting results of water vapor scale height data in different climate zones.

among different climate zones, with smaller amplitudes of the annual and semiannual cycle occurring in the subtropical monsoon climate zone. Specifically, the percentage of stations with amplitudes of annual cycle exceeding 0.2 km is 34.21 %, 21.05 %, 52.63 %, and 83.33 % for the subtropical monsoon climate, temperate monsoon climate, temperate continental climate, and high mountain plateau climate zones, respectively. For the amplitudes of semiannual cycle exceeding 0.09 km, the percentage becomes to 15.79 %, 78.95 %, 52.63 %, and 75.00 %.

To explore the relationship between the water vapor scale height and the meteorological parameters, including precipitable water vapor (PWV), temperature, water vapor pressure and pressure, the correlation coefficients were calculated and their spatial distributions were shown in Fig. 10. It showed that different climate zone have strong correlations with different meteorological parameters, e.g., the subtropical monsoon climate zone has the strongest correlation with PWV with an average correlation coefficient of 0.56; the temperate monsoon climate zone has a correlation with temperature and pressure, with an average correlation coefficient both reach to 0.44; the temperate continental climate zone and alpine plateau climate zone have the strongest correlation with temperature, with

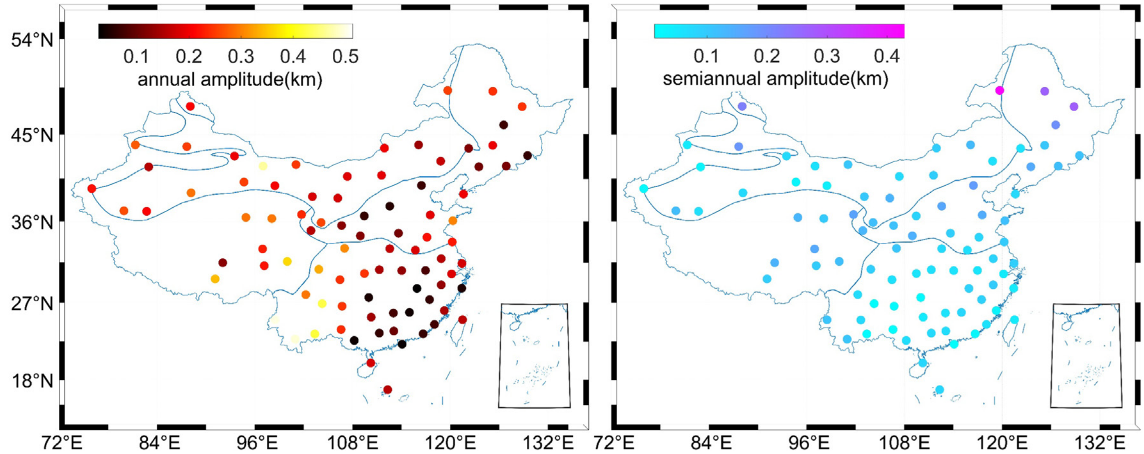

Fig. 9. Annual and semiannual amplitude of water vapor scale height in different climate zones.

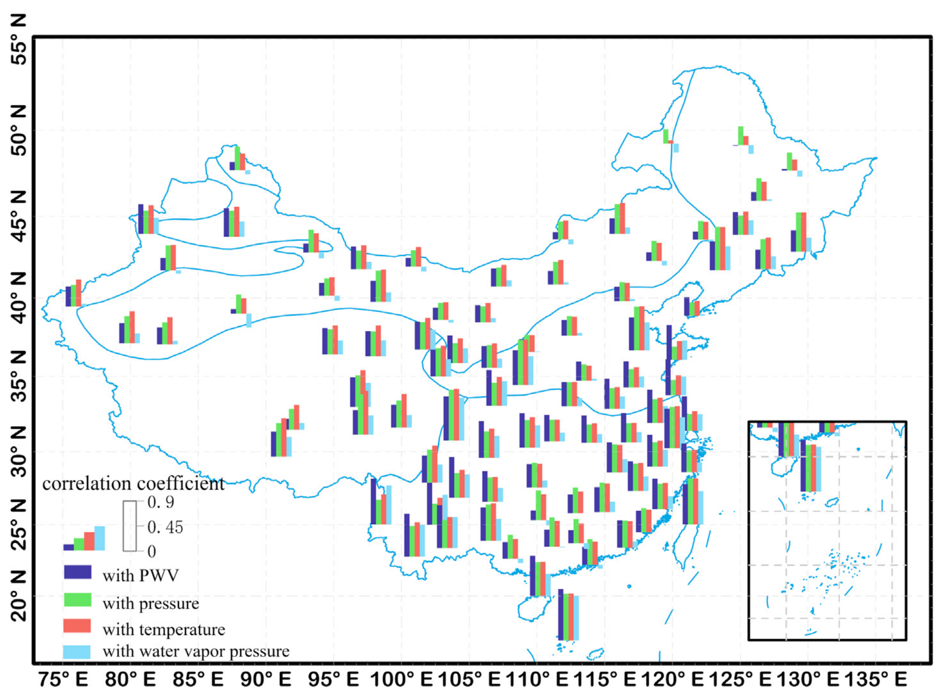

Fig. 10. The correlation coefficient histogram of water vapor scale height and meteorological parameters in different climate zones.

average correlation coefficients of 0.40 and 0.56, respectively. For all stations, the water vapor scale height exhibited positive correlations with PWV, temperature, and pressure. While for water vapor pressure, the negative correlation can be observed in some certain stations. Note that the correlation between water vapor scale height and water vapor pressure was generally weak, but strong relationship also can be observed in some stations with the correlation coefficients exceeding 0.8. It is concluded that the correlation between water vapor scale height and each meteorological parameters exhibits obvious geographical differences, and the meteorological parameter with strongest correlation with water vapor scale height is different in the four climate zones.

4. Conclusion

In this paper, we present a comprehensive analysis of long-term water vapor scale height using data from 89 radiosonde stations across China and dividing the whole country into four distinct climate zones, i.e., the subtropical monsoon climate zone, temperate monsoon climate zone, temperate continental climate zone and alpine plateau climate zone. It delves deep into the spatial–temporal variation of water vapor scale height, revealing that the temporal variation of water vapor scale height exhibits an annual periodicity, with certain stations also displaying a strong semiannual periodicity. The annual pattern features higher in summer and lower in winter, with a general trend of rising-then-decreasing. The value of water vapor scale height at nighttime is generally lower than that at daytime, with an average of 0.12 km. Among the climate zones, the subtropical monsoon climate zone exhibits the highest mean value of 2.45 km, while the temperate monsoon and the alpine plateau climate zone have lower mean values below 2.20 km. In addition, the subtropical monsoon and the alpine plateau climate zone exhibit a decline trend, whereas the temperate continental and the temperate monsoon climate zone show an increase trend. Across the country, the average change of water vapor scale height is 3.38 km per year. The relationship between water vapor scale height and various meteorological parameters displays distinct geographical variations, and the meteorological parameter demonstrating the strongest correlation with water vapor scale height varies across the four climate zones.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors would like to thank the University of Wyoming for providing the radiosonde data. This study is supported by the National Natural Science Foundation of China (42204022), the Open Research Fund of State Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University (23P02), the Fundamental Research Funds for the Central Universities (2024ZKPYDC02, 2023ZKPYDC10), Open Fund of Hubei Luojia Laboratory (230100031), China University of Mining and Technology-Beijing Innovation Training Program for College Students (202302014).

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